

Program: Perceptron Neural Network to Recognize Even and Odd Numbers

Important Theory for Practical Writing

Aim

To implement a Perceptron Neural Network to recognize even and odd numbers using ASCII values from 0 to 9.

Objective

To study the working of a Perceptron Neural Network and classify numbers into even and odd categories.

Short Important Theory (Write in Journal)

A Perceptron is the simplest type of Artificial Neural Network (ANN).

It works like a neuron of the human brain.

It takes input, applies weights and bias, and gives output using an activation function.

The Perceptron is mainly used for binary classification problems such as:

- Even/Odd detection
- Yes/No prediction
- Spam/Not Spam classification

In this practical, the perceptron is trained using numbers from 0 to 9.

Even numbers are represented by 0 and odd numbers by 1.

The model learns patterns during training and later predicts whether the entered number is even or odd.

Types of Perceptron

1. Single Layer Perceptron

- Contains only one neuron layer
- Used for linearly separable data
- Gives binary output

2. Multi Layer Perceptron

- Contains hidden layers
 - Solves complex problems
 - Works for linear and non-linear data
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Applications

1. Pattern recognition

2. Image classification
 3. Spam email detection
 4. Medical diagnosis
 5. Number classification
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Algorithm

1. Import required libraries
2. Prepare training dataset
3. Extract feature from input number

Code

```
import numpy as np

from sklearn.linear_model import Perceptron

import sys
```

```
data = [
    ('0', 48, 0),
    ('1', 49, 1),
    ('2', 50, 0),
    ('3', 51, 1),
    ('4', 52, 0),
    ('5', 53, 1),
    ('6', 54, 0),
    ('7', 55, 1),
    ('8', 56, 0),
    ('9', 57, 1),
]
```

```
def extract_feature(char_val):
    number = int(char_val)
    return [number % 2]
```

```
X_train = np.array([extract_feature(item[0]) for item in data])
```

```
y_train = np.array([item[2] for item in data])
```

```
perceptron = Perceptron(max_iter=100, random_state=0)
```

```
# Train the model
```

```
print("Training the Perceptron...")
```

```
perceptron.fit(X_train, y_train)
```

```
print("Training complete.\n")
```

```
def recognize_even_odd(char_input):
```

```
    if not char_input.isdigit():
```

```
        return f"Input '{char_input}' is not a valid digit."
```

```
    feature = extract_feature(char_input)
```

```
    prediction = perceptron.predict([feature])
```

```
    status = "Odd" if prediction[0] == 1 else "Even"
```

```
    return f"The number {char_input} is recognized as {status}."
```

```
print(recognize_even_odd('4'))
```

```
print(recognize_even_odd('7'))
```

```
print(recognize_even_odd('0'))
```

```
print(recognize_even_odd('9'))
```